

Quick Check

Lesson 3: Sines and Cosines of Complementary Angles: $\sin \theta = \cos(90^\circ - \theta)$

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 2.4: Trigonometry 1 | Lesson 3

Using mathematical tables, a pair finds that $\sin 25^\circ = 0.4226$. Which of these must also be equal to 0.4226?

A $\cos 25^\circ$

B $\cos 65^\circ$

C $\sin 65^\circ$

D $\tan 25^\circ$

Which pair of angles is complementary, so that the sine of one angle equals the cosine of the other?

A 20° and 160°

B 42° and 138°

C 35° and 65°

D 42° and 48°

In a right-angled triangle ABC, the right angle is at C, angle $A = \theta$ and angle $B = 90^\circ - \theta$. Side a is opposite angle A, side b is opposite angle B, and side c is the hypotenuse. Which ratio equals $\sin(90^\circ - \theta)$?

A b/c

B a/b

C b/a

D a/c

Which of these is equal to $\cos 70^\circ$?

A $\sin 20^\circ$

B $\sin 110^\circ$

C $\sin 35^\circ$

D $\sin 70^\circ$

Given that $\sin(2\theta + 10^\circ) = \cos(\theta + 20^\circ)$, what is the value of θ ?

A 30°

B 20°

C 10°

D 40°

Which statement about sine and cosine of complementary angles is correct?

A $\cos 65^\circ = \cos 25^\circ$

B $\sin 65^\circ = \cos 65^\circ$

C $\sin 65^\circ = \sin 25^\circ$

D $\sin 65^\circ = \cos 25^\circ$

A mast stands on level ground. From a point 40 m from its base, the angle of elevation to the top is 50° . In the right-angled triangle formed, the angle at the top of the mast is 40° . Why can an engineer use either the 50° angle or the 40° angle to work out the mast height?

- A** Because $\sin 50^\circ = \sin 40^\circ$ in the same triangle
- B** Because the tangent of 50° and the tangent of 40° are equal
- C** Because 50° and 40° are equal angles
- D** Because the two angles are complementary, so the sine of one equals the cosine of the other and both give the same triangle relationships

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).